

Effects of dietary flaxseed lignan extract on symptoms of benign prostatic hyperplasia.



A flaxseed lignan extract containing 33% secoisolariciresinol diglucoside (SDG) was evaluated for its ability to alleviate lower urinary tract symptoms (LUTS) in 87 subjects with benign prostatic hyperplasia (BPH). A randomized, double-blind, placebo-controlled clinical trial with repeated measurements was conducted over a 4-month period using treatment dosages of 0 (placebo), 300, or 600 mg/day SDG. After 4 months of treatment, 78 of the 87 subjects completed the study. For the 0, 300, and 600 mg/day SDG groups, respectively, the International Prostate Symptom Score (IPSS) decreased -3.67 ± 1.56 , -7.33 ± 1.18 , and -6.88 ± 1.43 (mean $\pm$ SE, $P = .100$, $< .001$, and $< .001$ compared to baseline), the Quality of Life score (QOL score) improved by -0.71 ± 0.23 , -1.48 ± 0.24 , and -1.75 ± 0.25 (mean $\pm$ SE, $P = .163$ and $.012$ compared to placebo and $P = .103$, $< .001$, and $< .001$ compared to baseline), and the number of subjects whose LUTS grade changed from "moderate/severe" to "mild" increased by three, six, and 10 ($P = .188$, $.032$, and $.012$ compared to baseline). Maximum urinary flows insignificantly increased 0.43 ± 1.57 , 1.86 ± 1.08 , and 2.7 ± 1.93 mL/second (mean $\pm$ SE, no statistical significance reached), and postvoiding urine volume decreased insignificantly by -29.4 ± 20.46 , -19.2 ± 16.91 , and -55.62 ± 36.45 mL (mean $\pm$ SE, no statistical significance reached). Plasma concentrations of secoisolariciresinol (SECO), enterodiol (ED), and enterolactone (EL) were significantly raised after the supplementation. The observed decreases in IPSS and QOL score were correlated with the concentrations of plasma total lignans, SECO, ED, and EL. In conclusion, dietary flaxseed lignan extract appreciably improves LUTS in BPH subjects, and the therapeutic efficacy appeared comparable to that of commonly used intervention agents of $\alpha 1A$ -adrenoceptor blockers and 5α -reductase inhibitors.